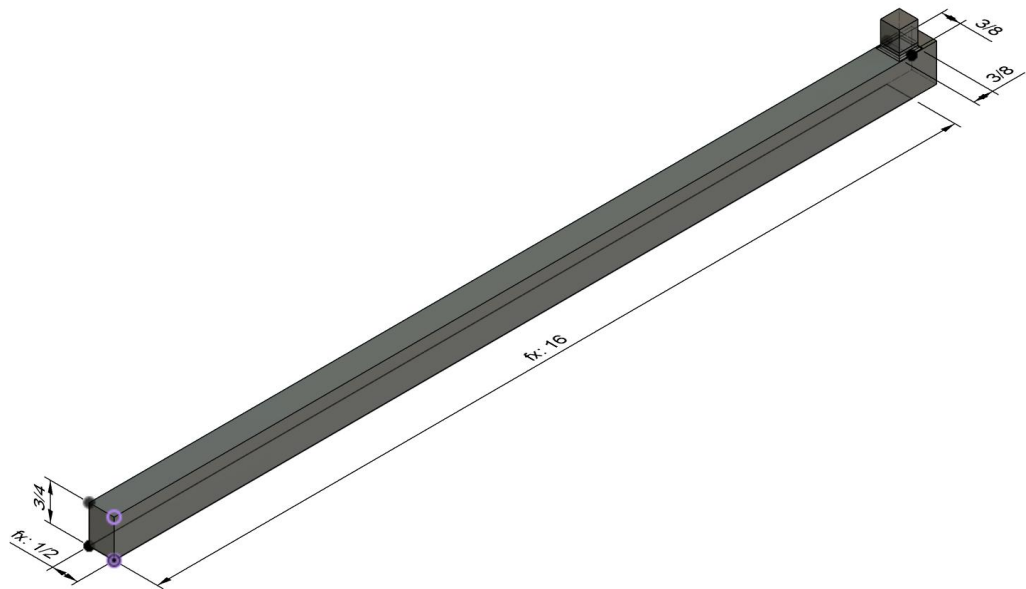


Part 2: Improved Design

1. Results

1.1. CAD Model Image



1.2. Material Selection

The material used is Ti-6AL-4V, an alloy of titanium with ideal material properties for this application. The main challenge of this design is having a sufficient strain at the location at the gauge while maintain wide factors of safety against failure. This alloy presents an ideal ratio of high yield strength (implying large factors of safety) and low elastic modulus (implying high strain). The relevant material properties are as follows:

$$\sigma_0 = 125 \text{ ksi}$$

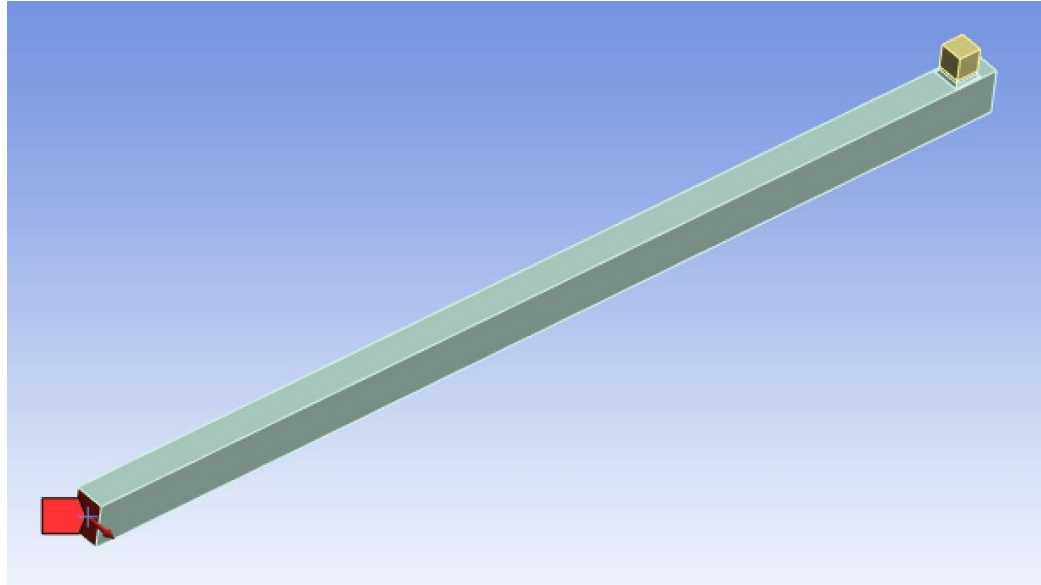
$$\sigma_N = 52 \text{ ksi (yield strength at } N = 10^6 \text{ cycles)}$$

$$K_{IC} = 76.4 \text{ ksi} \cdot \text{in}^{1/2}$$

$$E = 1.6 \cdot 10^4 \text{ ksi}$$

$$\nu = .31$$

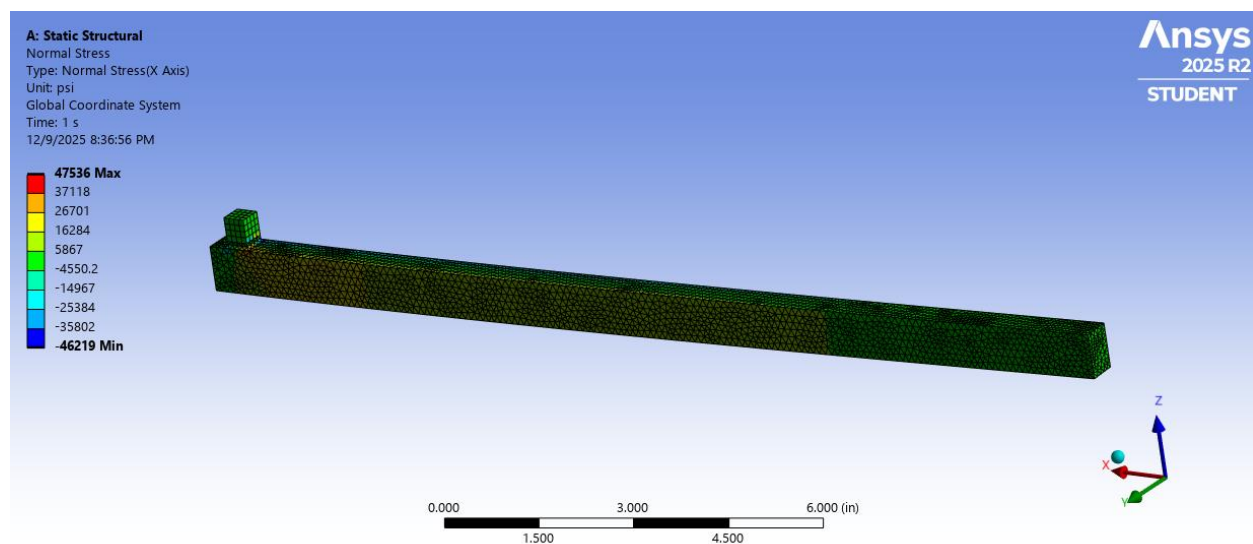
1.3. Loads and Boundary Conditions



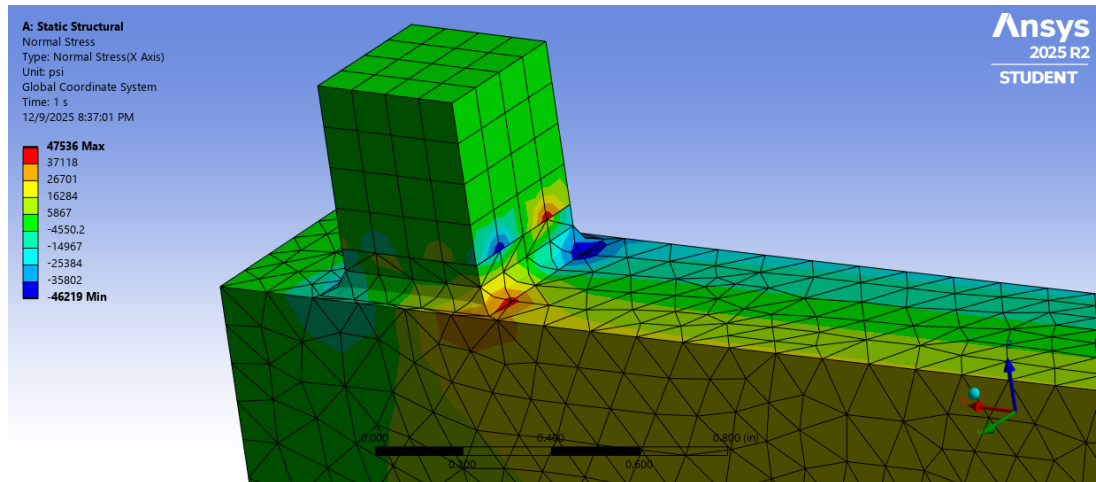
Loads: In red, a load of 37.5 lbs is applied horizontally (in the direction of the small red arrow) to apply a torque of 600 in · lbs to the drive.

Boundary conditions: The four faces on the outside of the drive are defined to have a displacement of zero in all directions. This defines the clamped condition on the upper .4" of the drive. The rest of the torque wrench is free.

1.4. Normal Strain Contours

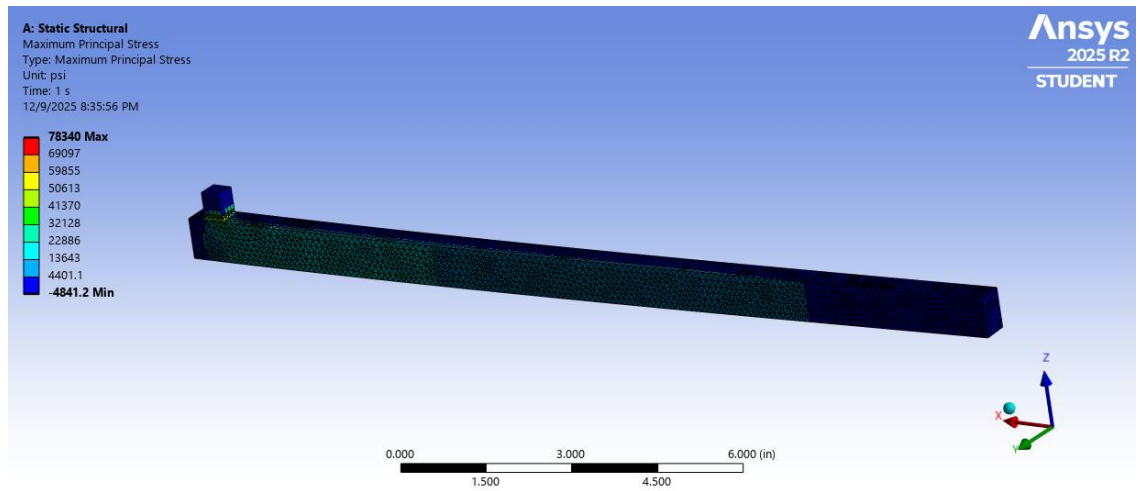


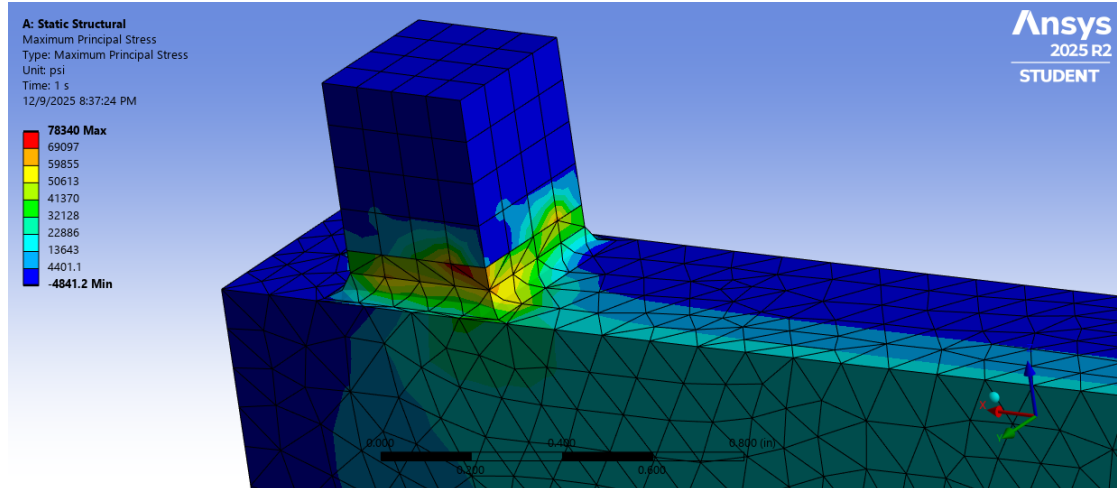
Plot of the normal stress in the x direction (pointing along the handle).



Locations of maximum and minimum stresses (stress concentration on the transition from the drive to the handle body)

1.5. Maximum principal stresses





1.6. Summary of Results

Maximum normal stress: 78.3 ksi

Load point deflection: .545 in

Strain at gauge locations: 1170, -1170 microstrain

1.7. Strain Gauge Sensitivity

The strain at the gauge location is 1170 microstrain. Since we are using a half bridge strain gauge, this implies a sensitivity of 1.170 mv/V.

1.8. Strain Gauge Selection

I selected a half-bridge strain gauge. One example is the SGT-Half-Bridge-Uniaxial strain gauge from DwyerOmega:

<https://www.dwyeromega.com/en-us/uniaxial-half-bridge-strain-gauges-with-transducer-quality/p/SGT-Half-Bridge-Uniaxial>

The dimensions of the product are .362"x.157", which can easily fit in the .75" profile of the torque wrench design.